

Specification for Floating Point Adder Simulation

In this assignment, you are required to design a floating point adder circuit which takes two floating points as inputs and provides their sum, another floating point as output. Each floating point will be 16 bits long with the following representation:

Sign	Exponent	Fraction
1 bit	5 bits	10 bits

- You have to implement your design in any simulator software of your choice. Please note that, you can directly use adder and subtractor (instead of an ALU) from the simulator wherever needed. Moreover, since construction of ALU is not the major focus of this assignment, you can take help from the Internet or other sources (or even use someone else's implementation) for the ALU part only.
- Implementations of **Overflow/Underflow** flags are necessary. **Rounding** and **normalization** is also necessary.
- It is advised that you prepare separate sub modules instead of one complex circuit.

Submission Guideline

- **Soft copy Submission:** Submit your assignment on Moodle. You have submit two files. 1) Block diagram of your design in pdf format, 2) simulation project files. Make a zip folder named Subsection No_Group No (e.g., B1_Group7). and submit this folder. Please ensure a single submission from each group.

Submission Deadline: For all lab groups: August 21, 11:55 PM

- **Report Submission:** Hard copy will be submitted on the respective lab time. The report should include the Block diagram and an example of step by step simulation of adding any two input numbers according to your design. During the evaluation, you have to demonstrate the process by showing intermediate results.
- For any query, please reach to farrihaafnan123@gmail.com.